while retaining a predefined delay–Doppler grid for the sparse channel representation. In practical propagation environments, significant power leakage caused by fractional Doppler shifts and fractional time delays can weaken the sparsity of the effective channel matrix [6]. This motivates learning effective-channel priors from data to capture channel structures beyond the assumed sparse representation [1], [6].

Score-based generative models learn channel priors from samples by estimating the score function of the channel distribution [10], [11]. In [6], score-based learning is performed directly on the AFDM effective channel to address fractional delay–Doppler off-grid leakage in channel estimation. For massive random access, [12] proposes a two-stage algorithm that first identifies active devices through a Transformer-based detector and then uses generative diffusion models for joint channel estimation and data detection. Both approaches rely on iterative posterior sampling, which requires repeated score function evaluations and incurs substantial computational cost [6], [12].

The score-based turbo message passing (STMP) algorithm in [1] incorporates generative channel priors for JADCE without iterative posterior sampling. Through Tweedie's formulas, separate first- and second-order score networks provide posterior means and variances for single-step minimum mean-squared error (MMSE) denoising under equivalent additive white Gaussian noise (AWGN) observations. However, this AWGN model is justified in the large-system limit with right-rotationally invariant random sensing matrices [1]. In the finite-dimensional, structured multi-frame AFDM system considered here, departures from these conditions may cause a mismatch between denoiser-input errors and the white Gaussian noise used for training. A similar limitation has been highlighted for structured compression matrices in [13]. This mismatch can degrade channel denoising accuracy and, consequently, JADCE performance. Moreover, [1] computes activity probabilities using Gaussian extrinsic messages obtained after projecting the channel posterior onto a Gaussian distribution. In general, matching posterior means and variances does not preserve the marginal likelihood associated with a non-Gaussian channel prior. Consequently, the resulting activity probabilities can be inaccurate, affecting the channel estimates exchanged with the linear estimation module.

Against this background, we investigate GNSS-independent AFDM-GFRA for LEO satellite IoT systems. Our main contributions are summarized as follows:

1) We propose a multi-frame AFDM transmission scheme that accommodates large differential delays and Doppler shifts at the satellite receiver. The scheme reduces modulation complexity through one-dimensional chirp modulation and uses a lower-dimensional effective-channel representation. By exploiting the quasi-static properties of the effective channel over consecutive frames, the scheme provides additional observations for JADCE.
2) We learn a score-based generative prior directly from multiantenna AFDM effective-channel samples with fractional delays and Doppler shifts, without imposing sparse or grid-aligned channel assumptions. To address the mismatch between denoiser-input errors and the white Gaussian noise used for training, we train the denoiser using colored Gaussian noise with different covariance matrices.
3) We develop a message-passing JADCE receiver for the multi-frame AFDM system. The receiver combines multi-frame linear estimation (MFLE), which suppresses inter-user interference, with generative-prior-aided joint denoising (GPJD), which uses the learned effective-channel prior for MMSE denoising. To reduce errors associated with a Gaussian approximation of the channel posterior, GPJD employs a learned density-ratio estimator to obtain posterior probabilities of activity. The two modules iteratively exchange Gaussian extrinsic messages to perform JADCE. Simulation results demonstrate that the proposed receiver improves activity detection and channel estimation performance compared with STMP across different signal-to-noise ratios and pilot-frame counts.

The remainder of this paper is organized as follows. Section II presents the system model and the multi-frame AFDM input–output relationship. Sections III–V describe the receiver design, the channel and activity denoisers, and the overall algorithm with its computational complexity analysis, respectively. Section VI presents the simulation results.

Notation: $\mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes a proper complex Gaussian law [14]. $\|\cdot\|_F$ denotes the Euclidean norm of all matrix or tensor entries; $(\cdot)^{\mathsf{H}}$ and $\mathrm{tr}(\cdot)$ denote conjugate transpose and trace.

II. System Model

We consider a LEO satellite-enabled grant-free random access system, where AFDM is adopted to mitigate the doubly dispersive effect inherent in terrestrial-satellite links (TSLs) [3], [6]. The system comprises K single-antenna terrestrial devices that communicate with a LEO satellite equipped with a uniform planar array (UPA) consisting of $M = M_y M_z$ antennas. During each transmission block, active devices share the same resources for pilot transmission to the satellite. These terrestrial devices operate without GNSS capabilities and do not pre-compensate for delay and Doppler shifts [3].

A. Terrestrial-Satellite Link Model

The TSLs experience fast variations due to the high mobility of the LEO satellite. Hence, this time variation and multipath propagation require a doubly dispersive channel model [3]. We assume that the terrestrial devices are stationary and all paths from the kth device have a common Doppler shift ν_k [6]. Within a transmission block, the path gains, delays, Doppler shifts, and angles of arrival remain approximately constant. We set $t_{\mathrm{c}} = 0$ at the beginning of the transmission block. After delay compensation for τ_c at the satellite receiver, the channel impulse response from the kth device to the satellite at time instant t_{c} can be represented as

$$\mathbf{h}_k(t_{\mathrm{c}}, \tau) = \mathbf{a}_k \sum_{\ell=1}^{L_k} g_{k\ell} e^{j2\pi\nu_k t_{\mathrm{c}}} \delta(\tau - \tau_{k\ell}), \tag{1}$$

where $k = 1, \ldots, K$, L_k is the number of physical paths, $g_{k\ell}$ is the path gain of the ℓth path, and $\delta(\cdot)$ is the Dirac delta

function. Here $\tau_{k\ell} = \widetilde{\tau}_{k\ell} - \tau_c$ represents the differential delay for the ℓth path from the kth device to the satellite, where $\widetilde{\tau}_{k\ell}$ and τ_c are denoted as propagation delay and common delay, respectively. The common delay corresponds to the propagation time between the satellite and the closest point on Earth within the beam coverage [3].

The array response vector $\mathbf{a}_k$ at the satellite side can be written as [3], [15]

$$[\mathbf{a}_k]_{n_z+M_z n_y} = e^{j2\pi(d_y n_y \mu_{y,k} + d_z n_z \mu_{z,k})/\lambda}, \tag{2}$$

where $n_z = 0, \ldots, M_z - 1$, $n_y = 0, \ldots, M_y - 1$, λ is the wavelength, and d_v is the spacing between adjacent antennas along the v-axis with $v \in \{y, z\}$. Here $\mu_{z,k}$ and $\mu_{y,k}$ are the directional cosines along the z-axis and y-axis, respectively.

B. Transmission Scheme

In the proposed scheme, each active device transmits pilot symbols over Q consecutive AFDM frames within one transmission block. AFDM modulation and demodulation are performed independently for each frame, and the received signals are collected for JADCE. A total of N chirp subcarriers are shared among the devices for pilot transmission [1], [6]. Let $P_{qkn} \in \mathbb{C}$ be the pilot symbol transmitted by the k-th device during the q-th AFDM frame on the n-th chirp subcarrier. We construct a pilot vector as $\mathbf{p}_{qk} = [P_{qk0}, \ldots, P_{qk,N-1}]^{\mathsf{T}} \in \mathbb{C}^N$, $k = 1, \ldots, K$, $q = 0, \ldots, Q-1$. The pilot symbols are known at the satellite receiver. Let $\mathbf{P} \in \mathbb{C}^{Q \times KN}$ denote the pilot matrix, with entries $[\mathbf{P}]_{q,(k-1)N+n} = P_{qkn}$.

We impose the condition that $\mathbf{P}\mathbf{P}^{\mathsf{H}} = KE_p\mathbf{I}_Q$, where $\|\mathbf{p}_{qk}\|^2 = E_p$ for all k and q [1]. The design of $\mathbf{P}$ is deferred to Section VI-A.

The time-domain transmitted signal is generated through the inverse discrete affine Fourier transform (IDAFT) applied to $\mathbf{p}_{qk}$, mathematically formulated as [8]

$$s_{qk}[t] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} P_{qkn} e^{j2\pi(c_1 t^2 + c_2 n^2 + nt/N)}, \tag{3}$$

where $t = 0, \ldots, N-1$, and c_1 and c_2 are AFDM modulation parameters. The signal in (3) can be represented in matrix form as [8]

$$\mathbf{s}_{qk} = \mathbf{F}_{\mathrm{A}}^{\mathsf{H}} \mathbf{p}_{qk},$$

where

$$\mathbf{F}_{\mathrm{A}} = \left[\frac{1}{\sqrt{N}} e^{-j2\pi(c_2 r^2 + rt/N + c_1 t^2)} \right]_{r,t=0}^{N-1} \tag{4}$$

denotes the discrete affine Fourier transform (DAFT) matrix. Here $T_s = 1/(N\Delta f)$ gives the sampling interval, with Δf denoting the subcarrier spacing.

Assume that the chirp-periodic prefix (CPP) length is greater than or equal to the maximum differential delay, i.e., $N_{\mathrm{CPP}} T_s \geq \max_{k,\ell} \tau_{k\ell}$. The CPP is given by [9]

$$\begin{aligned} s_{qk}[t] &= e^{j2\pi c_1 (t^2 - \langle t \rangle_N^2)} s_{qk}[\langle t \rangle_N], \\ &-N_{\mathrm{CPP}} \leq t < 0, \end{aligned} \tag{5}$$

where $\langle x \rangle_N = x - N\lfloor x/N \rfloor$ and $2Nc_1 \in \mathbb{N}_+$. Each AFDM frame spans a duration of $T_{\mathrm{frame}} = (N + N_{\mathrm{CPP}})T_s$.

To account for fractional delays, we define the transmit waveform for the nth chirp subcarrier at time ξT_s as

$$\varphi_n(\xi) = \frac{1}{\sqrt{N}} e^{j2\pi[c_2 n^2 + c_1 \xi^2 + n w_\xi/N - \zeta_n(w_\xi) w_\xi]}, \tag{6}$$

where $w_\xi = \langle \xi \rangle_N$ and $\zeta_n(w) = \lfloor 2c_1 w + n/N \rfloor$. Note that $\varphi_n(t) = [\mathbf{F}_{\mathrm{A}}^{\mathsf{H}}]_{tn}$ for $t = 0, \ldots, N-1$, while $\varphi_n(\xi)$ satisfies (5) for $\xi < 0$.

Denote by $H_k^{(q)} \in \mathbb{C}^{M \times N \times N}$ the effective channel from the kth device to the satellite during the qth AFDM frame, after removing the CPP and applying the DAFT. For $q = 0$, the effective channel $H_k = H_k^{(0)}$ can be expressed as

$$H_{kmrn} = [\mathbf{a}_k]_m \sum_{\ell=1}^{L_k} g_{k\ell} [\mathbf{G}(\tau_{k\ell}/T_s, \nu_k; N_{\mathrm{CPP}})]_{rn}, \tag{7}$$

where n and r correspond to the transmitted and received symbols, respectively, and $m = 0, \ldots, M-1$. For a path with gain 1, delay dT_s, and Doppler shift ν, the channel response can be expressed in a matrix form as

$$\begin{aligned} \mathbf{G}(d, \nu; g) &= \mathbf{F}_{\mathrm{A}} \mathbf{D}_{\nu,g} \mathbf{B}_d, \\ \mathbf{D}_{\nu,g} &= \mathrm{diag}\{e^{j2\pi\nu T_s (g+t)}\}_{t=0}^{N-1}, \\ [\mathbf{B}_d]_{tn} &= \varphi_n(t-d), \end{aligned} \tag{8}$$

where $t_{\mathrm{c}} = (g+t)T_s$ for $t = 0, \ldots, N-1$ after removing the CPP, and $g = N_{\mathrm{CPP}}$ for frame 0. Please refer to Appendix B for the detailed derivation and the expression for integer d.

Based on (8) with $g = g_q = N_{\mathrm{CPP}} + q(N + N_{\mathrm{CPP}})$, we can derive the effective channel in the qth AFDM frame as

$$\begin{aligned} H_k^{(q)} &= e^{jq\beta_k} H_k, \\ \beta_k &= 2\pi \left(\nu_k T_{\mathrm{frame}} - \left\lfloor \nu_k T_{\mathrm{frame}} + \frac{1}{2} \right\rfloor \right), \end{aligned} \tag{9}$$

where $\beta_k \in [-\pi, \pi)$. The phase rotation $e^{jq\beta_k}$ accumulates across multiple frames, while H_k remains approximately constant over Q consecutive frames. The pilot symbols including this phase rotation are given by

$$A_{qkn} = P_{qkn} e^{jq\beta_k}. \tag{10}$$

C. Problem Formulation

Each device determines whether to access the channel in an independent and identically distributed (i.i.d.) manner. We use an indicator function α_k to model the device activity as [1]

$$\alpha_k = \begin{cases} 1, & \text{if the } k\text{th device is active,} \\ 0, & \text{otherwise.} \end{cases}$$

Let ρ represent the probability of a device being active, i.e., $\Pr(\alpha_k = 1) = \rho$, where $\rho \ll 1$. As a result, only a small subset of the devices is active in any given transmission block. The activity states are assumed to remain constant during one block transmission. Let $X_k = \alpha_k H_k$.

After removing the CPP and applying the DAFT, the received signal $\mathbf{Y}_q \in \mathbb{C}^{M \times N}$ during the qth AFDM frame is given by [1]

$$Y_{qmr} = [\mathbf{Y}_q]_{mr} = \sum_{k=1}^{K} \sum_{n=0}^{N-1} A_{qkn} X_{kmrn} + W_{qmr}, \tag{11}$$

where $\mathbf{W}_q = [W_{qmr}]_{m,r} \in \mathbb{C}^{M\times N}$ is an AWGN matrix with the elements i.i.d. drawn from $\mathcal{CN}(0,\sigma^2)$.

We express the received signal $\mathbf{Y}$ over the Q AFDM frames in a more compact form as

$$\mathbf{Y} = \mathbf{A}\mathbf{X} + \mathbf{W}, \tag{12}$$

where $\mathbf{Y}, \mathbf{W} \in \mathbb{C}^{Q\times MN}$, $\mathbf{A} \in \mathbb{C}^{Q\times KN}$, and $\mathbf{X} \in \mathbb{C}^{KN\times MN}$. The matrices $\mathbf{Y}$, $\mathbf{W}$, $\mathbf{A}$, and $\mathbf{X}$ have the following entries:

$$\begin{aligned} [\mathbf{Y}]_{q,mN+r} &= Y_{qmr}, \\ [\mathbf{W}]_{q,mN+r} &= W_{qmr}, \\ [\mathbf{A}]_{q,(k-1)N+n} &= A_{qkn}, \\ [\mathbf{X}]_{(k-1)N+n,mN+r} &= X_{kmrn}. \end{aligned} \tag{13}$$

Since the Q frames share the same $\mathbf{X}$, additional pilot frames provide more observations without increasing the number of channel coefficients to be estimated. Due to the sporadic access of the devices, $\mathbf{X}$ exhibits a block-sparsity structure. Specifically, if the kth device is inactive, we have $X_k = \alpha_k H_k = \mathbf{0}$ [1]. For $Q < KN$, the inverse problem in (12) is underdetermined.

In this paper, we aim to simultaneously detect the active devices and estimate the corresponding channels based on the observation $\mathbf{Y}$ under a Bayesian inference framework [1]. Power leakage arises from fractional delays and Doppler shifts, which can weaken the sparsity of H_k [6]. We therefore estimate the full H_k without imposing channel sparsity. We use the distribution of the training channels as the channel prior $q_{\mathrm{tr}}(H_k)$, without factorizing this prior element-wise. We assume that H_k and α_k are independent for different values of k, and that H_k is independent of α_k. With $\widehat{\mathbf{A}}$ and $\widehat{\sigma}^2$ fixed as in Section V-B, we factorize the joint posterior distribution according to Bayes' theorem and the conditional independence relationships among the variables, expressed as [1]

$$\begin{aligned} &q(\mathbf{X}, \{H_k\}_{k=1}^K, \{\alpha_k\}_{k=1}^K \mid \mathbf{Y}) \\ &\quad\propto \prod_{j=1}^{MN} p(\mathbf{y}_j \mid \mathbf{x}_j) \\ &\quad\quad\times \prod_{k=1}^{K} p(X_k \mid H_k, \alpha_k) q_{\mathrm{tr}}(H_k) p(\alpha_k), \end{aligned} \tag{14}$$

where $\mathbf{x}_j \in \mathbb{C}^{KN}$ and $\mathbf{y}_j \in \mathbb{C}^Q$ denote the jth column of $\mathbf{X}$ and $\mathbf{Y}$, respectively. Here $p(\mathbf{y}_j \mid \mathbf{x}_j) = \mathcal{CN}(\mathbf{y}_j; \widehat{\mathbf{A}}\mathbf{x}_j, \widehat{\sigma}^2\mathbf{I}_Q)$, $p(X_k \mid H_k, \alpha_k)$ enforces $X_k = \alpha_k H_k$, and $p(\alpha_k) = \rho^{\alpha_k}(1-\rho)^{1-\alpha_k}$. The MMSE estimate of X_k and the posterior probability of activity for the kth device are given by

$$\begin{aligned} X_k^{\mathrm{MMSE}} &= \mathbb{E}_q[X_k \mid \mathbf{Y}], \\ p_k^{\mathrm{Bayes}} &= q(\alpha_k = 1 \mid \mathbf{Y}). \end{aligned} \tag{15}$$

III. Bayesian Receiver Design

The posterior in (14) couples the channels of all devices through the multi-frame observations. Computing the estimates in (15) therefore requires marginalization over both channel coefficients and activity states. We develop a message-passing receiver that separates the linear observation model from the channel and activity priors. The resulting local posterior reveals the channel moments and activity probabilities that must be computed from the full channel prior.

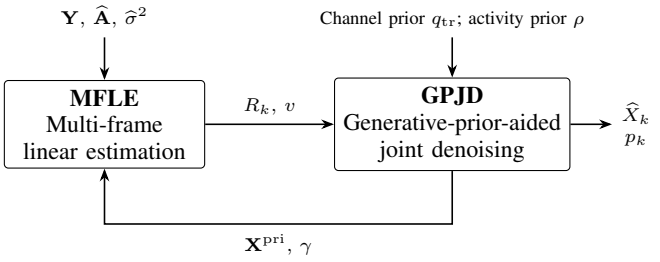


Fig. 1. Information exchanged between the MFLE and GPJD modules.

A. Posterior Factorization and Message Passing

The two products in (14) have different roles. The likelihood factors couple devices within each observation column, whereas the prior factors associate each activity variable with the entire channel tensor of that device. Following turbo message passing [1], [16], we group these calculations into multi-frame linear estimation (MFLE) and generative-prior-aided joint denoising (GPJD), respectively. MFLE uses the received pilots to update the channel estimates; GPJD incorporates the channel and activity priors into these estimates. This grouping retains the dependence among the coefficients of each H_k in q_{tr}.

The two modules exchange Gaussian messages, as illustrated in Fig. 1. In each module, the incoming message is combined with the corresponding likelihood or prior, and the resulting posterior is approximated by matching its mean and average variance. Dividing this Gaussian approximation by the incoming message gives the extrinsic message to the other module, as in expectation propagation [1], [17]. Both the estimate and its uncertainty thus enter the next update. We first derive the linear update and then determine the posterior moments required by joint denoising.

B. Multi-Frame Linear Estimation

Let the incoming Gaussian message have mean $\mathbf{X}^{\mathrm{pri}} \in \mathbb{C}^{KN\times MN}$ and per-column covariance $\gamma\mathbf{I}_{KN}$, where $\gamma > 0$. Multiplying this message by the Gaussian likelihood in (14) gives the linear posterior mean and covariance [1, Sec. III-A]:

$$\mathbf{X}_{\mathrm{L}} = \mathbf{X}^{\mathrm{pri}} + \gamma\widehat{\mathbf{A}}^{\mathsf{H}}(\widehat{\sigma}^2\mathbf{I}_Q + \gamma\widehat{\mathbf{A}}\widehat{\mathbf{A}}^{\mathsf{H}})^{-1}(\mathbf{Y} - \widehat{\mathbf{A}}\mathbf{X}^{\mathrm{pri}}), \tag{16}$$

$$\mathbf{\Sigma}_{\mathrm{L}} = (\gamma^{-1}\mathbf{I}_{KN} + \widehat{\sigma}^{-2}\widehat{\mathbf{A}}^{\mathsf{H}}\widehat{\mathbf{A}})^{-1}, \qquad t_{\mathrm{L}} = \frac{\mathrm{tr}\,\mathbf{\Sigma}_{\mathrm{L}}}{KN}. \tag{17}$$

The device-dependent inter-frame phases enter through $\widehat{\mathbf{A}}$; their factors $e^{j(q-q')\beta_k}$ can prevent pilot row orthogonality from carrying over to this matrix. We therefore retain its full Gram matrix in the posterior covariance.

To obtain a scalar-variance message, we replace $\mathbf{\Sigma}_{\mathrm{L}}$ by $t_{\mathrm{L}}\mathbf{I}_{KN}$, preserving its trace. Dividing the resulting Gaussian approximation by the incoming message yields [16], [17]

$$v_{\mathrm{L}} = (t_{\mathrm{L}}^{-1} - \gamma^{-1})^{-1}, \qquad \mathbf{R}_{\mathrm{L}} = v_{\mathrm{L}}\left(\frac{\mathbf{X}_{\mathrm{L}}}{t_{\mathrm{L}}} - \frac{\mathbf{X}^{\mathrm{pri}}}{\gamma}\right). \tag{18}$$